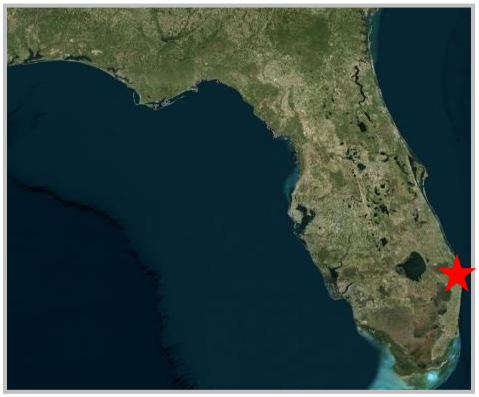
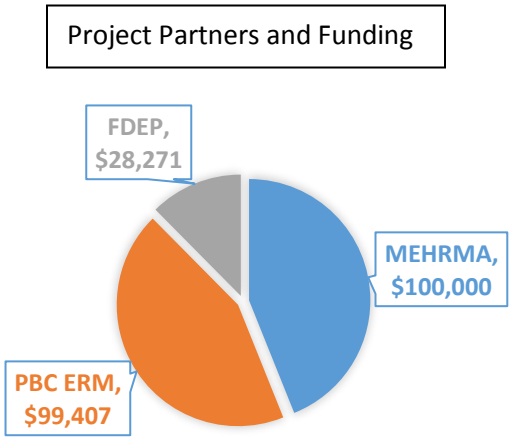
Jewell Cove Shoreline Restoration, Lake Worth Lagoon, Palm Beach County, FL

FY 15/16





Project Description



Figure 1: Volunteers planted red mangroves and Spartina. July 27, 2016.

Jewell Cove is an 11.44 acre area that lies within the City of Lake Worth along the eastern shoreline of the Lake Worth Lagoon. The project area included an eroding shoreline, a maritime hammock dominated by exotics, and submerged lands. The purpose of this project was to remove exotic vegetation, protect the shoreline from erosion, and install native plants along 1,100 linear feet of shoreline. Limestone boulders, proven to be viable oyster reef habitat in this area, were installed to protect the shoreline and provide mangrove, *Spartina,* and oyster habitats. The boulders also add shoreline resilience in regards to climate change and sea level rise and enhance recreational fishing opportunities. Future components of the project, a shell rock pedestrian pathway through the hardwood hammock and an informational kiosk, will add public access and educational opportunities.

Jewell Cove is part of a bigger project known as Old Bridge Park. Additional components of the Old Bridge Park project to which this funding did not contribute include repairing a failing seawall and installation of a shade shelter and day-use dock for the public for recreation and fishing. Kiosks and signs about the overall project will be installed to provide information to the public about the environmental benefits and the sponsors of the project. The Old Bridge Park project received additional funding from the Florida Inland Navigation District and from the FDEP through the 2014-2015 General Appropriations Act program grant. The funding provided by MEHRMA for the Jewell Cove Shoreline portion of the project was used as match towards the FIND and FDEP grants for the Old Bridge Park project. When Old Bridge Park is complete, the total project costs (Old Bridge Park components and Jewell Cove components) are estimated to be approximately $764,000.

Benefits

- Attenuate waves and boat wakes
- Reduce turbidity over submerged lands in the project area
- Provide substrate and appropriate hydrology for mangroves and salt marsh plants
- Provide substrate for the recruitment and colonization of oysters, sponges, and algae
- Provide nursery and foraging habitat for a variety of wildlife including juvenile fish, invertebrates, oysters, wading birds, manatees and juvenile green sea turtles.

FWC Strategic Plan Elements Supported

Theme I, Goal 2: Ensure sufficient habitats exist to support healthy and diverse fish and wildlife populations.

Strategies:

- Protect lands and waters critical to sustaining healthy and diverse fish and wildlife populations through diverse incentive programs.
- Manage habitats to sustain healthy and diverse fish and wildlife populations.

Additionally, this project is of regional importance and is highlighted in the 2013 Lake Worth Lagoon Management Plan and is recognized as a priority project by the Lake Worth Lagoon Initiative.

Acreage restored by habitat type (footprint)

Total- 2.34 acres oyster, mangrove, saltmarsh and tropical hardwood hammock:

- Breakwater/oyster habitat- 0.28 acre
- Mangroves- 0.45 acre
- *Spartina* and open shoreline- 0.78 acre
- Tropical hardwood hammock- 0.83 acre

Acreage affected by restoration activities: 11.44 acres (includes acreage listed above and 9.1 acres of submerged lands)



Figure 2: Jewel Cove shoreline prior to construction. North of project site viewed to the south. April 21, 2016.



Figure 3: Newly planted shoreline with constructed limerock breakwater. July 27, 2016.